

Part H solution :

(15) The receiver only gets the quantized error signal $x - \hat{q}[n] = x^n[n] + e - \hat{q}[n]$ which means it needs its own copy of the predictor to generate $x^n[n]$ from previously reconstructed samples. Without matching predictor, the receiver has no way to turn the error signal back into the actual waveform.

(16) if the predictors differ even slightly, the $x^n[n]$ computed at the transmitter (used to form $e[n]$) won't match the $x^n[n]$ computed at the receiver (used to reconstruct $x - \hat{q}[n]$). This mismatch introduces an additional, uncontrolled error beyond the quantization error, causing the reconstructed signal to drift away from the original. The whole DPCM scheme relies on both sides tracking the same prediction.

(17)

The receiver never has access to the original samples $x[n]$ - it only ever has the reconstructed samples $x_q[n]$. If the transmitter based its prediction on the true $x[n-1]$, its $\hat{x}[n]$ would differ from what the receiver can compute (since the receiver only has $x_q[n-1]$). By using $x_q[n-1]$ at the transmitter too, both sides work from identical information keeping the predictors synchronized.

(18)

~~What if~~ This causes a mismatch between transmitter and receiver, and the prediction errors don't cancel correctly during reconstruction. Small discrepancies can accumulate over time (error propagation/drift), leading to a growing reconstruction error that isn't due to quantization at all - potentially causing the receiver's estimate of the signal to diverge significantly from the true signal.

(19) Speech signals are highly correlated from one sample to next one (they change relatively slowly within short time windows due to the physical constraints of vocal tract movement). This ~~strong~~ strong sample-to-sample correlation means a predictor can estimate $x[n]$ quite accurately from past samples. So the prediction error $e[n]$ has much smaller amplitude / variance than $x[n]$ itself allowing it to be quantized with fewer bits for same quality, i.e. DPCM exploits redundancy that's abundant in speech.

(20) No, if the samples are completely uncorrelated knowing $x[n-1]$ gives no information about $x[n]$. The best predictor would essentially give $x[n] \approx 0$ (or constant) meaning $e[n] \approx x[n]$ - the prediction error has the same power / range as the original signal. In this case DPCM provides no dynamic range reduction, a net no coding gain over straight PCM; the predictor simply cannot exploit correlation that isn't there.